BMC Sports Science, Medicine and Rehabilitation

RESEARCH

Open Access

Associations of ankle strength and range of motion with reactive strength performance in elite youth soccer players

Recep Fatih Kayhan[1], Halil İbrahim Çakır[2], Esranur Terzi[3], Ömer Aksoy[4] and Serdar Bayrakdaroğlu[5*]

Abstract

Background Ankle dorsiflexion and plantar flexion are fundamental movements that contribute to lower-limb stability and explosive performance in sports. Adequate ankle strength and active range of motion (AROM) may be associated with athletic performance, whereas limitations in these parameters may be associated with impaired function and increased injury risk. Therefore, this study aimed to examine the associations between ankle strength, AROM, and reactive strength index (RSI) in youth soccer players.

Methodology A total of 79 male youth soccer players from the U15, U16, U17, and U19 age categories participated in the study. Isometric ankle strength, ankle ROM, and RSI were assessed, and their associations were examined using forced-entry multiple linear regression models, in which dominant- and non-dominant-leg dorsiflexion and plantar flexion values were entered simultaneously as predictors of RSI outcomes, as predictors of RSI outcomes, with Benjamini–Hochberg false discovery rate (FDR) correction applied to account for multiple comparisons.

Results In the dominant leg (DL), both dorsiflexion and plantar flexion AROM were associated with DL RSI, explaining 18.0% and 15.5% of the variance, respectively. In the non-dominant leg (NDL), plantar flexion AROM showed a significant association with NDL RSI (17.4% of variance), whereas the association with dorsiflexion AROM (5.3%) did not remain significant after correction for multiple comparisons. For two-leg RSI, DL dorsiflexion AROM, DL plantar flexion AROM, and NDL plantar flexion AROM explained 9.6%, 7.4%, and 8.6% of the variance, respectively. Regarding strength variables, DL dorsiflexion and plantar flexion strength explained 8.9% and 12.9% of the variance in DL RSI, whereas NDL dorsiflexion strength explained 23.3% of the variance in NDL RSI. For two-leg RSI, DL dorsiflexion strength, DL plantar flexion strength, and NDL dorsiflexion strength explained 12.3%, 18.2%, and 14.8% of the variance, respectively. Overall, DL plantar flexion strength and NDL dorsiflexion strength showed the strongest associations with RSI outcomes.

Conclusions These findings suggest that ankle ROM and strength may be associated with reactive strength performance in young soccer players, although the explained variance was generally low-to-moderate across models (R^2 = 5.3–23.3%). In particular, dominant-leg plantar flexion strength and non-dominant-leg dorsiflexion strength appear to be relevant contributors to RSI outcomes. Accordingly, targeted mobility and strength exercises for the ankle joint may help support reactive strength performance.

*Correspondence:
Serdar Bayrakdaroğlu
bayrakdaroglu85@gmail.com

Full list of author information is available at the end of the article

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

Keywords Dorsiflexion, Explosive power, Lower limb biomechanics, Plantar flexion, Reactive strength index, Soccer performance

Introduction

In the field of sports sciences, understanding the performance parameters of young athletes and examining the relationships between these parameters are of great importance both to support athletic development and to minimize the risk of injury. Recent evidence in elite youth soccer players has also demonstrated associations between functional movement quality and physical performance measures, including jumping, sprinting, and agility [1]. The ankle is a fundamental joint that plays an important role in both static and dynamic movements, and the functionality of this joint is among the factors that may be associated with athletic success [2]. Ankle dorsiflexion and plantar flexion strength, range of motion (ROM), and reactive strength performance are important parameters related to both stability and explosive power capacity in athletes [3].

Ankle dorsiflexion and plantar flexion strength are critical factor in achieving optimal biomechanical alignment during movement [4]. In the literature, insufficient strength or imbalance has been reported to be associated with Achilles tendinopathy, ankle sprain, and other lower-extremity injuries in athletes [5, 6]. Recent evidence in professional soccer players has further highlighted the importance of ankle function by demonstrating persistent functional and balance deficits in players with a history of lateral ankle sprain [7]. ROM, as a parameter reflecting flexibility and movement capacity, may contribute to performance and help reduce injury risk [8]. Dorsiflexion ROM is particularly important in activities requiring high performance, such as rapid change of direction and sprinting in soccer players [9]. Reactive strength performance is another important indicator used to measure explosive power capacity and neuromuscular efficiency in athletes [10, 11]. Reactive strength index (RSI) is typically calculated as the ratio of jump height to ground contact time during plyometric tasks such as drop jumps and is widely used as an indicator of explosive strength and stretch–shortening cycle performance [12]. In the literature, it has been reported that a low RSI increases the risk of injury in athletes and may negatively affect their performance [10, 13].

Ankle dorsiflexion and plantar flexion strength and ROM may be associated with RSI because ankle function contributes to rapid force generation and efficient force transmission during reactive movements [14, 15]. In particular, adequate plantar flexor strength and dorsiflexion ROM may support effective force production during rapid ground-contact tasks [16, 17]. Therefore, ankle strength and ROM may be important factors associated with reactive strength performance.

Although inter-limb asymmetries between the dominant and non-dominant leg have been increasingly studied in relation to strength and jump performance in youth soccer players [18, 19], most of this evidence has focused on general strength or jump-height asymmetries, and comparatively little is known about how dominant- and non-dominant-leg ankle strength and ROM are differentially associated with both unilateral and bilateral reactive strength performance. Clarifying this distinction may have direct practical relevance for individualized training prescription, as it would indicate whether mobility and strength interventions should be limb-specific rather than applied uniformly to both legs. Previous studies on lower extremity performance parameters in youth soccer players have predominantly focused on age- and maturation-related differences [20–22]. Recent evidence has further linked movement quality and explosive performance measures to jumping and agility outcomes in youth athletes [1, 23, 24]. However, studies comprehensively examining the interrelationships among these parameters are still scarce [3]. In particular, understanding the associations between ankle strength, ROM, and reactive strength performance may help address this gap in the literature. Accordingly, the primary aim of this study was to examine the associations between ankle dorsiflexion and plantar flexion strength, ROM, and reactive strength performance in youth soccer players. The secondary aim was to examine differences in these parameters across age categories (U15, U16, U17, and U19). Based on the mechanical role of ankle mobility and strength in the stretch–shortening cycle (SSC), we hypothesized that (i) greater ankle dorsiflexion and plantar flexion ROM would be positively associated with ipsilateral single-leg RSI and two-leg RSI, reflecting the capacity of a larger ROM to enhance eccentric energy storage during the amortization phase of ground contact; and (ii) greater ankle dorsiflexion and plantar flexion strength would be positively associated with ipsilateral single-leg RSI and two-leg RSI, reflecting the contribution of ankle musculature to efficient force transmission during the concentric phase of the SSC. Given the differential loading patterns typically imposed on the dominant and non-dominant limbs during sport-specific actions such as kicking and cutting, we further expected the dominant leg to show stronger associations for propulsion-related parameters (e.g., plantar flexion strength), whereas the non-dominant leg, which is more frequently engaged in postural stabilization, was expected to show stronger associations

for control-related parameters (e.g., dorsiflexion strength and ROM).

Materials and methods

Participants

The study included young male youth soccer players competing in the U15 ($n = 19$), U16 ($n = 20$), U17 ($n = 19$), and U19 ($n = 21$) age categories. Based on a priori power analysis (G*Power 3.1; $f^2 = 0.56$, $\alpha = 0.05$, $1-\beta = 0.85$), the minimum required sample size was calculated as 39 participants. Accordingly, a total of 79 athletes were included in the study, all of whom were actively training and competing at the time of data collection. Data collection took place during the in-season period. All participants were competing at the national level within their respective age categories.

The inclusion criteria were as follows: (i) no acute or chronic lower extremity pain, (ii) no lower extremity injury within the past six months, (iii) being a registered competitive soccer player with a valid license, (iv) at least three years of active playing experience, and (v) participation in at least two training sessions per week.

The study was approved by the Ethics Committee of Recep Tayyip Erdoğan University (Approval No: 2025/376; Date: 21/05/2025), and all procedures were conducted in accordance with the Declaration of Helsinki. Prior to participation, written assent was obtained from all participants after the study procedures were explained to them, and written informed consent was obtained from their legal guardians. Permission was also obtained from the affiliated sports clubs.

General test procedure

Prior to testing, the athletes were provided with detailed information regarding the purpose, significance, and procedures of the assessments. All tests were conducted in a laboratory setting between 15:00 and 17:00, with participants wearing standard sports attire. To ensure standardization, participants were instructed to refrain from any strenuous physical activity for 24 h prior to testing. Anthropometric measurements, including height, body mass, and dominant side, were recorded. Subsequently, the athletes performed a standardized 5-minute warm-up on a cycle ergometer. The testing order was identical for all participants and was not randomized. Within each testing session, the order of limb (dominant/non-dominant) and movement direction (dorsiflexion/plantar flexion) assessments was standardized and applied identically to all participants, rather than randomized. All assessments were completed on the same day. Following the warm-up, testing was performed in a fixed order for all participants: ankle strength measurements were completed first, followed by ROM assessment, and finally the RSI (drop jump) protocol. A minimum of 5 min of rest was provided between each testing block to minimize the potential effect of fatigue from the preceding test. Performance testing commenced immediately following the warm-up, and all measurements were conducted in accordance with standardized protocols.

Muscle strength measurement protocol

Ankle muscle strength was assessed using a handheld dynamometer (ActivForce 2, ActivForce, California, USA). The device has been reported to demonstrate high reliability, with Cronbach's alpha values ranging between 0.867 and 0.969 [25]. Participants were seated on a standard-height chair without armrests, with the hips and knees flexed at 90°. One foot was placed flat on the ground, while the heel of the tested limb remained in contact with the floor [26]. To ensure familiarization with the testing protocol; participants performed four submaximal isometric contractions in both dorsiflexion and plantar flexion using both dominant and non-dominant limbs. These trials were not included in the analysis. Following familiarization, a total of eight maximal isometric contractions were performed across both limbs, with two contractions performed in each direction (dorsiflexion and plantar flexion) for each leg (i.e., four contractions per leg). The highest peak isometric force value (kg) obtained for each direction and limb was recorded for analysis. A one-minute rest interval was provided between trials to minimize the effects of fatigue [27], consistent with previous handheld dynamometry protocols indicating that inter-trial rest periods of this duration are sufficient to prevent a cumulative fatigue effect across repeated maximal isometric contractions.

During dorsiflexion measurements, participants were instructed to dorsiflex the ankle to the maximum range and maintain the position while gradually increasing force to a maximal level. The dynamometer was positioned on the dorsal surface of the metatarsal heads, and maximal contraction was sustained for at least five seconds [4]. For plantar flexion measurements, the same protocol was applied, with the dynamometer placed on the plantar surface of the metatarsal heads [28]. The dynamometer was manually stabilized by the same examiner throughout all trials to ensure consistent force application across participants.

Strength values were recorded in kilograms (kg) as output by the device software; where relevant, these values can be converted to newtons (N) by multiplying by 9.81. Trials in which participants were unable to maintain maximal contraction for at least five seconds were deemed invalid and repeated. Standardized verbal encouragement was provided during every maximal contraction to ensure maximal voluntary effort. Standardized verbal encouragement was provided during every maximal contraction to ensure maximal voluntary effort.

AROM measurement protocol

Ankle dorsiflexion and plantar flexion active range of motion (AROM) were assessed for both limbs using a standard goniometer. Measurements were performed with participants in a supine position, with the knees fully extended. The axis of the goniometer was aligned with the lateral malleolus, the stationary arm was positioned along the longitudinal axis of the fibula, and the movable arm was aligned with the fifth metatarsal. Participants were instructed to remain relaxed and to actively move the ankle through its full ROM during the assessment. Each movement (dorsiflexion and plantar flexion) was performed three times for each limb, and the mean value was used for statistical analysis. All measurements were conducted by the same researcher following a standardized protocol to minimize intra-rater variability [29].

RSI measurement protocol

Vertical reactive strength index (RSI) was assessed using a jump mat system (Jump Mat Pro, FSL JumpMat, Tyrone, Ireland). RSI was calculated as the ratio of jump height to ground contact time (RSI = jump height / contact time), automatically computed by the jump mat system's software. The device has been reported to demonstrate medium reliability, with coefficients of 0.64 for drop jump [30]. Participants performed unilateral drop jumps from a 30-cm box onto the jump mat. Immediately after landing, they performed a maximal vertical jump with minimal ground contact time while keeping their hands on their hips throughout the movement. A 30 cm box height was selected based on previous validation studies indicating that this height elicits a fast stretch-shortening cycle response with a short ground contact time, appropriate for assessing reactive strength in youth athletes without imposing excessive eccentric loading [30]. Trials were performed separately for both the dominant and non-dominant limbs. The dominant limb was defined as the preferred kicking leg, determined by self-report. Movements such as excessive knee flexion during flight or failure to land properly on the mat were considered invalid, and the trial was repeated. Each participant completed two valid trials for each limb, with a two-minute rest interval between trials. RSI values (m/s) were recorded for analysis, and the higher of the two valid trials was used for analysis.

Higher RSI values are indicative of greater reactive strength and more efficient stretch–shortening cycle performance.

Statistical analysis

An a priori power analysis was conducted using G*Power (version 3.1; Heinrich Heine University, Düsseldorf, Germany) to determine the minimum required sample size. The "Linear multiple regression: Fixed model, R^2 increase" option within the F-test family was selected. All statistical analyses were performed using SPSS (version 27.0; IBM Corp., Armonk, NY, USA). Descriptive statistics are presented as mean ± standard deviation (SD) for all variables. Differences in anthropometric characteristics (height, body mass, body fat percentage), ankle AROM, strength, and RSI outcomes across age categories (U15, U16, U17, U19) were examined using one-way analysis of variance (ANOVA). Prior to regression analysis, the normality of the RSI data was assessed using skewness and kurtosis values, which were found to be within acceptable ranges, indicating approximate normal distribution. Multicollinearity among independent variables was evaluated using the variance inflation factor (VIF), with values below 10 indicating no significant multicollinearity. Heteroscedasticity was assessed using the Breusch–Pagan test, with a p-value greater than 0.05 indicating homoscedasticity of residuals. Autocorrelation was examined using the Durbin–Watson statistic, with values between 1.5 and 2.5 considered acceptable. In addition, residual plots were inspected to verify the assumptions of normality, linearity, and independence of errors. These assumptions (normality, multicollinearity, heteroscedasticity, autocorrelation) were evaluated separately for each of the four regression models [31].

For each outcome (single-leg and two-leg RSI), simple (bivariate) linear regression models were conducted separately for each predictor to obtain individual unstandardized (B) and standardized (β) regression coefficients, standard errors, t-values, p-values, 95% confidence intervals, and R^2 values. In addition, for each outcome, a combined simultaneous (forced-entry) multiple linear regression model -including all conceptually related dominant- and non-dominant-leg predictors together, without a stepwise or hierarchical selection procedure- was fitted to obtain the overall model F-statistic, model-level p-value, and adjusted R^2, which are reported alongside the individual predictor coefficients in Tables 2 and 3. Statistical significance was set at $\alpha = 0.05$. Given that 16 predictor coefficients were tested across Tables 2 and 3, Benjamini–Hochberg false discovery rate (FDR) correction was applied to control the family-wise error rate, with FDR-adjusted p-values reported alongside uncorrected p-values in each table. As an additional exploratory analysis addressing a reviewer's recommendation, ROM × strength interaction effects and dominant–non-dominant limb asymmetry indices -calculated as [(Dominant – Non-dominant) / mean (Dominant, Non-dominant)] × 100- were tested as predictors of RSI outcomes, with FDR correction applied separately across these eight exploratory tests (Table 4).

Table 1 Descriptive statistics of anthropometric characteristics, AROM and RSI levels of football players

Variable	U15 (*n* = 19)	U16 (*n* = 20)	U17 (*n* = 19)	U19 (*n* = 21)	ANOVA
	M ± SD	M ± SD	M ± SD	M ± SD	*p*
Height (cm)	168.42 ± 5.07	173.15 ± 4.76	175.58 ± 4.94	178.62 ± 5.49	< 0.001
Weight (kg)	58.26 ± 5.69	63.25 ± 5.89	65.55 ± 5.72	69.32 ± 7.18	< 0.001
Body fat %	10.13 ± 1.45	10.37 ± 1.38	9.10 ± 3.23	11.99 ± 3.66	0.010
DL Dorsal AROM	19.26 ± 3.83	16.50 ± 3.90	21.00 ± 4.51	19.62 ± 4.79	0.013
NDL Dorsal AROM	19.58 ± 2.36	16.65 ± 3.01	21.37 ± 4.88	21.24 ± 2.84	< 0.001
DL Plantar AROM	45.37 ± 6.51	40.45 ± 6.66	36.26 ± 5.46	41.00 ± 5.15	< 0.001
NDL Plantar AROM	45.74 ± 7.29	41.45 ± 8.46	40.74 ± 6.56	38.86 ± 7.77	0.041
DL Dorsal Strength	17.81 ± 2.87	16.13 ± 2.58	18.51 ± 3.72	17.10 ± 2.34	0.076
NDL Dorsal Strength	17.76 ± 3.50	15.96 ± 2.48	18.03 ± 2.97	16.75 ± 2.42	0.097
DL Plantar Strength	29.07 ± 5.03	26.39 ± 4.82	32.63 ± 5.50	30.17 ± 5.18	0.003
NDL Plantar Strength	27.29 ± 6.12	29.49 ± 4.93	33.15 ± 7.03	30.64 ± 3.58	0.015
Two Leg RSI	1.40 ± 0.14	1.23 ± 0.22	1.28 ± 0.32	1.31 ± 0.35	0.239
DL RSI	1.20 ± 0.13	1.09 ± 0.23	1.11 ± 0.23	1.10 ± 0.15	0.209
NDL RSI	1.19 ± 0.11	1.05 ± 0.22	1.10 ± 0.22	1.10 ± 0.13	0.119

DL Dominant leg, *NDL* Non-dominant leg, *AROM* Active range of motion, *RSI* Reactive strength index, *cm* Centimeter; *%* Percent, *kg* Kilogram, *n* Sample size, *M* Mean, *SD* Standard deviation

Table 2 Regression results for the association between DL and NDL AROM and single and two-leg RSI

Predictor Variable	Outcome Variable	B	SE	(β)	t	*p*	FDR-adj *p* (BH)	95% CI Lower	95% CI Upper	R^2	F	R^2	Adjusted R^2	*P*
DL Dorsal AROM	DL RSI	0.018	0.004	0.425	4.116	< 0.001**	0.001	0.009	0.027	0.180	13.064	0.256	0.236	< 0.001**
DL Plantar AROM		0.011	0.003	0.394	3.756	< 0.001**	0.001	0.005	0.017	0.155				
NDL Dorsal AROM	NDL RSI	0.011	0.005	0.230	2.072	0.042*	0.051	0.004	0.021	0.053	10.725	0.220	0.200	< 0.001**
NDL Plantar AROM		0.010	0.002	0.417	4.031	< 0.001**	0.001	0.005	0.014	0.174				
DL Dorsal AROM	DL-NDL (Two-Legs) RSI	0.020	0.007	0.328	3.047	0.003*	0.006	0.003	0.030	0.096	3.853	0.172	0.128	0.007*
DL Plantar AROM		0.012	0.004	0.293	2.688	0.009*	0.012	0.004	0.016	0.074				
NDL Dorsal AROM		0.008	0.008	0.117	1.031	0.306	0.326	-0.017	0.019	0.001				
NDL Plantar AROM		0.011	0.004	0.313	2.889	0.005*	0.008	0.002	0.015	0.086				

DL Dominant leg, *NDL* Non-dominant leg, *AROM* Active range of motion, *RSI* Reactive strength index, *SE* Standard error, *BH* Benjamini-Hochberg, *CI* Confidence interval

*$p < 0.05$, **$p < 0.001$

Results

Table 1 shows that anthropometric characteristics (height, body mass, and body fat percentage) differed significantly across age categories (all $p \leq 0.010$), reflecting expected growth- and maturation-related trends. Ankle AROM values also varied significantly across age groups for all four AROM variables (all $p < 0.05$), with values generally lower in the U16 group compared to the other categories. In terms of plantar AROM, the highest values were found in the U15 group, while a decrease was observed in the U16 group followed by an increase in the U17–U19 groups. In contrast, ankle strength differed significantly across age groups for plantar flexion strength (both DL and NDL, $p \leq 0.015$) but not for dorsiflexion strength (both DL and NDL, $p > 0.05$), and RSI outcomes did not differ significantly across age categories (all $p > 0.05$).

Direct paired comparisons between the dominant and non-dominant leg showed no statistically significant differences in ROM, strength, or RSI values (all $p > 0.05$; Supplementary Table S1).

Multiple linear regression analyses were used to examine the associations between ankle ROM and single-leg RSI (Table 2). The regression model including DL dorsiflexion and plantar flexion ROM and DL RSI was statistically significant ($F_{(2,76)} = 13.064$, $p < 0.001$, $R^2 = 0.256$, adjusted $R^2 = 0.236$). DL dorsiflexion ROM was positively associated with DL RSI (B = 0.018, SE = 0.004, β = 0.425, 95% CI [0.009, 0.027], $p < 0.001$), explaining 18.0% of the variance, while DL plantar flexion ROM was also positively associated with DL RSI (B = 0.011, SE = 0.003, β = 0.394, 95% CI [0.005, 0.017], $p < 0.001$), explaining 15.5% of the variance (Fig. 1a).

Similarly, the regression model including NDL dorsiflexion and plantar flexion ROM and NDL RSI was statistically significant ($F_{(2,76)} = 10.725$, $p < 0.001$, $R^2 = 0.220$, adjusted $R^2 = 0.200$). NDL dorsiflexion ROM explained 5.3% of the variance in NDL RSI (B = 0.011, SE = 0.005, β = 0.230, 95% CI [0.004, 0.021], $p = 0.042$), whereas NDL

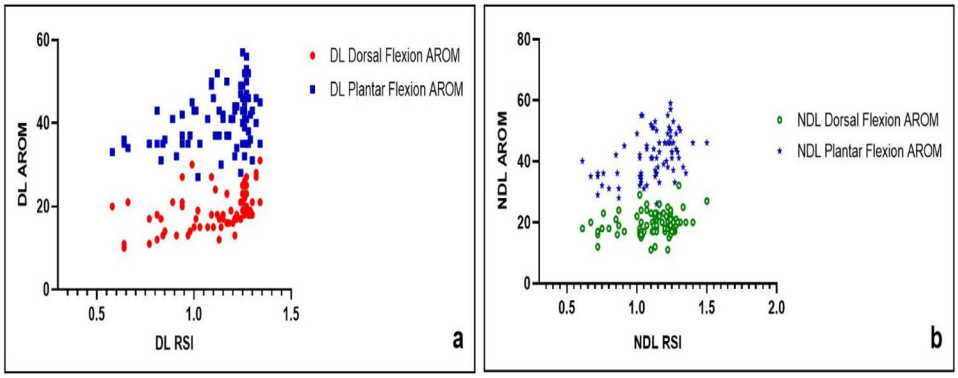


Fig. 1 Effect of DL and NDL dorsal and plantar flexion AROM on single leg RSI: (**a**) DL; (**b**) NDL

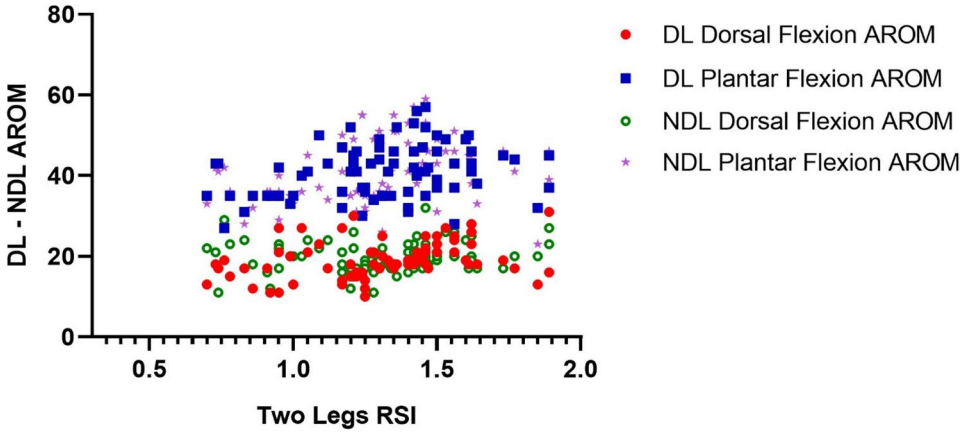


Fig. 2 Effect of DL-NDL dorsal and plantar flexion AROM on two leg RSI

plantar flexion ROM explained 17.4% of the variance (B = 0.010, SE = 0.002, β = 0.417, 95% CI [0.005, 0.014], $p < 0.001$) (Fig. 1b). Following Benjamini-Hochberg FDR correction for multiple comparisons (16 tests), the association between NDL dorsal ROM and NDL RSI no longer reached statistical significance (p_FDR = 0.051), whereas all other associations in this table remained significant.

The same analytical approach was applied to examine the association between ankle ROM and two-leg RSI (Table 2). The regression model including DL dorsiflexion, DL plantar flexion, NDL dorsiflexion, and NDL plantar flexion ROM and two-leg RSI was statistically significant ($F_{(4,74)} = 3.853$, $p = 0.007$, $R^2 = 0.172$, adjusted $R^2 = 0.128$). DL dorsiflexion ROM explained 9.6% of the variance in two-leg RSI (B = 0.020, SE = 0.007, β = 0.328, 95% CI [0.003, 0.030], $p = 0.003$), DL plantar flexion ROM explained 7.4% (B = 0.012, SE = 0.004, β = 0.293, 95% CI [0.004, 0.016], $p = 0.009$), and NDL plantar flexion ROM explained 8.6% (B = 0.011, SE = 0.004, β = 0.313, 95% CI [0.002, 0.015], $p = 0.005$) (Fig. 2). NDL dorsiflexion ROM was not significantly associated with two-leg RSI (B = 0.008, SE = 0.008, β = 0.117, 95% CI [-0.017, 0.019], $p = 0.306$). All associations that were significant at the uncorrected level remained significant after FDR correction; non-significant associations remained non-significant.

An identical modeling approach was used to examine the associations between ankle strength and single-leg RSI (Table 3). The regression model including DL dorsiflexion and plantar flexion strength and DL RSI was statistically significant ($F_{(2,76)} = 5.753$, $p = 0.005$, $R^2 = 0.131$, adjusted $R^2 = 0.109$). DL dorsiflexion strength explained 8.9% of the variance in DL RSI (B = 0.019, SE = 0.007, β = 0.298, 95% CI [0.005, 0.033], $p = 0.008$), while DL plantar flexion strength explained 12.9% (B = 0.012, SE = 0.004, β = 0.359, 95% CI [0.005, 0.020], $p = 0.001$) (Fig. 3a).

Similarly, the regression model including NDL dorsiflexion and plantar flexion strength and NDL RSI was statistically significant ($F_{(2,76)} = 12.394$, $p = 0.001$, $R^2 = 0.246$, adjusted $R^2 = 0.223$). NDL dorsiflexion strength explained 23.3% of the variance in NDL RSI (B = 0.030, SE = 0.006, β = 0.483, 95% CI [0.018, 0.042], $p = 0.001$), whereas NDL plantar flexion strength showed no significant association (B = 0.001, SE = 0.004, β = 0.030, 95% CI [-0.006, 0.008], $p = 0.796$) (Fig. 3b). All associations that were significant at the uncorrected level remained

Table 3 Regression results for the association between DL and NDL strength and single and two-leg RSI

Predictor Variable	Outcome Variable	B	SE	(β)	t	*p*	FDR-adj *p* (BH)	95% CI Lower	95% CI Upper	R^2	F	R^2	Adjusted R^2	*p*
DL Dorsal Strength	DL RSI	0.019	0.007	0.298	2.741	0.008*	0.011	0.005	0.033	0.089	5.753	0.131	0.109	0.005**
DL Plantar Strength		0.012	0.004	0.359	3.373	0.001*	0.002	0.005	0.020	0.129				
NDL Dorsal Strength	NDL RSI	0.030	0.006	0.483	4.841	< 0.001**	< 0.001	0.018	0.042	0.233	12.394	0.246	0.226	0.001*
NDL Plantar Strength		0.001	0.004	0.030	0.259	0.796	0.796	-0.006	0.008	0.001				
DL Dorsal Strength	DL-NDL (Two-Legs) RSI	0.034	0.010	0.366	3.453	0.001*	0.002	0.014	0.053	0.123	5.084	0.216	0.173	0.001*
DL Plantar Strength		0.022	0.005	0.439	4.289	< 0.001**	< 0.001	0.002	0.032	0.182				
NDL Dorsal Strength		0.036	0.010	0.384	3.652	< 0.001**	0.001	0.010	0.047	0.148				
NDL Plantar Strength		0.008	0.005	0.165	1.469	0.146	0.167	-0.010	0.011	0.015				

DL Dominant leg, *NDL* non-dominant leg, *RSI* Reactive strength index, *SE* Standard error, *BH* Benjamini-Hochberg, *CI* Confidence interval

*$p < 0.05$, **$p < 0.001$

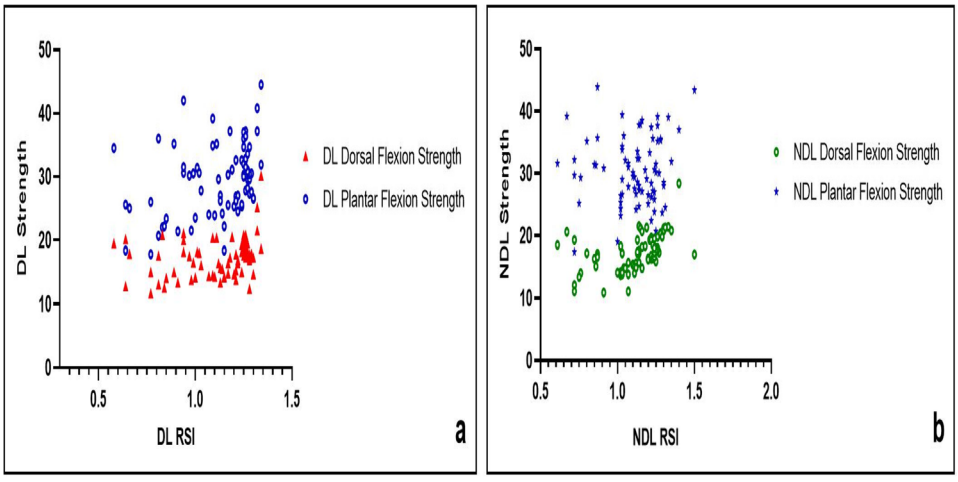


Fig. 3 Effect of DL and NDL dorsal and plantar flexion strength on single leg RSI: (**a**) DL; (**b**) NDL

significant after FDR correction; non-significant associations remained non-significant.

Finally, the same approach was applied to examine the association between ankle strength and two-leg RSI (Table 3). The regression model including DL dorsiflexion, DL plantar flexion, NDL dorsiflexion, and NDL plantar flexion strength and two-leg RSI was statistically significant ($F_{(4,74)} = 5.084$, $p = 0.001$, $R^2 = 0.216$, adjusted $R^2 = 0.173$). DL dorsiflexion strength explained 12.3% of the variance in two-leg RSI (B = 0.034, SE = 0.010, β = 0.366, 95% CI [0.014, 0.053], $p = 0.001$), DL plantar flexion strength explained 18.2% (B = 0.022, SE = 0.005, β = 0.439, 95% CI [0.002, 0.032], $p < 0.001$), and NDL dorsiflexion strength explained 14.8% (B = 0.036, SE = 0.010, β = 0.384, 95% CI [0.010, 0.047], $p = 0.001$) (Fig. 4). NDL plantar flexion strength was not significantly associated with two-leg RSI (B = 0.008, SE = 0.005, β = 0.165, 95% CI [-0.010, 0.011], $p = 0.146$). All associations that were significant at the uncorrected level remained significant after FDR correction; non-significant associations remained non-significant. Across all regression models, the proportion of variance explained by ankle AROM and strength ranged from 0.1% to 25.6%, indicating that these variables account for only a limited, and in some cases negligible, portion of the overall variability in RSI performance.

Exploratory analyses testing ROM × strength interactions and dominant–non-dominant limb asymmetry as predictors of RSI are presented in Table 4. None of these associations remained significant after FDR correction (all p_FDR > 0.05), although two associations (DL plantar flexion AROM × strength interaction, $p = 0.033$; dorsal AROM asymmetry, $p = 0.043$) reached significance at the uncorrected level.

Discussion

In this study, the associations between ankle strength, AROM, and reactive strength performance were investigated. The results indicate that ankle AROM and strength are associated with reactive strength performance at various levels. In general, a steady increase in ankle AROM and strength values is not observed with increasing age, and significant decreases occur in some age groups. This suggests that the growth and development process varies

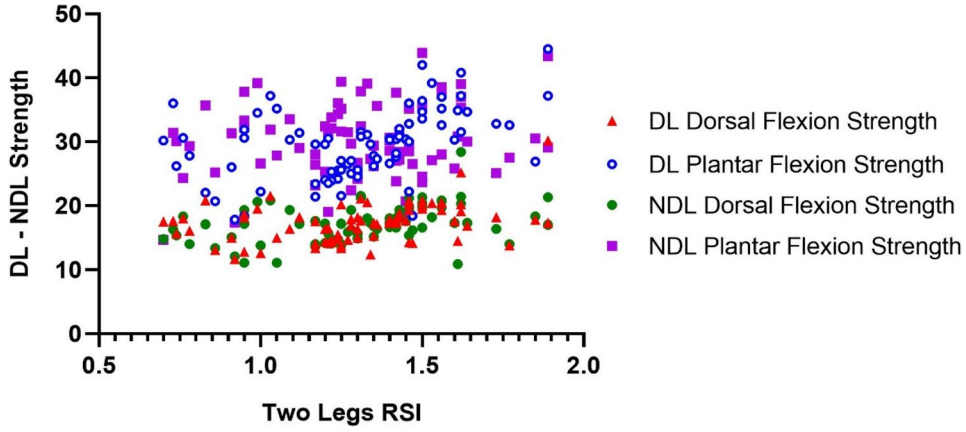


Fig. 4 Effect of DL-NDL dorsal and plantar flexion strength on two leg RSI

Table 4 Exploratory analysis of ROM × strength interaction effects and dominant–non-dominant limb asymmetry as predictors of RSI

Predictor / Interaction Term	Outcome Variable	B	SE	(β)	t	*p*	95% CI Lower	95% CI Upper	R^2	F	*p*	*p* (FDR)
ROM × Strength Interaction Effects												
DL Dorsal AROM × DL Dorsal Strength	DL RSI	-0.001	0.001	-0.041	-0.568	0.572	-0.003	0.001	0.184	5.636	0.002*	0.667
DL Plantar AROM × DL Plantar Strength	DL RSI	-0.001	0.001	-0.282	-2.173	0.033*	-0.003	< 0.001	0.279	9.670	< 0.001**	0.173
NDL Dorsal AROM × NDL Dorsal Strength	NDL RSI	-0.001	0.002	-0.057	-0.493	0.623	-0.005	0.003	0.242	7.961	< 0.001**	0.667
NDL Plantar AROM × NDL Plantar Strength	NDL RSI	< 0.001	< 0.001	0.076	0.731	0.467	-0.001	0.001	0.190	5.851	0.001*	0.667
Dominant–Non-Dominant Limb Asymmetry Indices												
Dorsal AROM Asymmetry Index	Two-Leg RSI	0.003	0.001	0.228	2.055	0.043*	< 0.001	0.005	0.052	4.224	0.043*	0.173
Plantar AROM Asymmetry Index	Two-Leg RSI	-0.001	0.002	-0.049	-0.431	0.667	-0.004	0.003	0.002	0.186	0.667	0.667
Dorsal Strength Asymmetry Index	Two-Leg RSI	-0.001	0.003	-0.054	-0.477	0.635	-0.007	0.004	0.003	0.227	0.635	0.667
Plantar Strength Asymmetry Index	Two-Leg RSI	0.002	0.001	0.200	1.794	0.077	<-0.001	0.005	0.040	3.219	0.077	0.205

DL Dominant leg, *NDL* Non-dominant leg, *AROM* Active range of motion, *RSI* Reactive strength index, *SE* Standard error, *CI* Confidence interval, *FDR* Benjamini–Hochberg false discovery rate-adjusted p-value across all eight exploratory tests

*$p < 0.05$, **$p < 0.00$

depending on individual differences and that training loads may play an important role in the development of strength and AROM.

Ankle AROM and RSI performance

The results suggest that ankle AROM is associated with RSI, with dorsiflexion AROM showing a stronger association in the dominant leg and plantar flexion AROM showing a stronger association in the non-dominant leg. This asymmetry may suggest that the dominant leg contributes more to propulsion, whereas the non-dominant leg is more involved in stabilization; however, the association between NDL dorsiflexion AROM and NDL RSI did not survive correction for multiple comparisons (FDR-adjusted $p = 0.051$) and should therefore be interpreted with caution. For two-leg RSI, dominant-leg dorsiflexion AROM showed the strongest association among the ROM variables, suggesting that the dorsiflexion capacity of the dominant leg may play a particularly relevant role in two-legged jumps. Additionally, the stronger association of non-dominant-leg plantar flexion AROM with two-leg RSI compared to single-leg RSI may reflect the involvement of this limb in postural stabilization during bilateral movements.

Studies have shown that ankle and knee joint ROM plays a potentially important role in youth soccer players achieving high RSI values in movements such as cutting, sprinting, and jumping [32]. A larger ROM allows the muscle-tendon unit to stretch more and store elastic energy, which can translate into a faster and stronger

concentric contraction [32, 33]. Accordingly, it appears that the levels of strength produced in specific ROM regions may influence athletes' elastic energy utilization and rapid strength production [8]. In research, the efficiency of the transition from eccentric loading to concentric strength production (stretch-shortening cycle) stands out as a determining factor on the reactive strength capacity of athletes [10]. In this process, the mechanical properties of the muscle-tendon unit are of great importance [9] and the effect of ROM is inevitable in movements involving a stretch-shortening cycle. In this study, the change in RSI values with increasing ankle ROM is thought to be related to the efficiency of this transition process [13]. A wide ROM may help to store more energy in the eccentric phase and subsequently utilize this energy more efficiently [8]. Especially in athletes with low ankle ROM, it has been shown that ground contact time is prolonged and RSI values are lower [15, 34]. This suggests that failure to achieve optimal muscle-tendon tension during eccentric loading may negatively affect reactive strength production. A wider dorsiflexion angle may allow the eccentric contraction phase to occur more effectively at the moment of ground contact, and this may be associated with higher RSI values [10]. In the existing literature, there are many studies examining the effect of ROM on muscle strength and neuromuscular adaptations. In particular, it is stated that training with full ROM provides greater strength gains compared to partial ROM [35]. In a study by Clark et al. [8] the effects of resistance training at different ROM levels on athletes' reactive strength performance were examined and it was found that mixed ROM training provided greater improvement compared to full ROM training. In this context, low ankle ROM in youth soccer players may lead to suboptimal muscle-tendon tension during eccentric loading and thus low RSI. Therefore, the ROM levels of athletes should be taken into account in the design of training programs and optimal ROM should be targeted to improve reactive strength performance.

Ankle strength and RSI performance

The results show that ankle strength is associated with RSI. In the DL, plantar flexion strength showed a stronger association with RSI than dorsiflexion strength, suggesting that plantar flexors may be more relevant to the propulsive strength of the dominant leg. In the NDL, only dorsiflexion strength was significantly associated with RSI, which may indicate that the non-dominant leg is more involved in stabilization and balance, with dorsiflexion strength potentially playing a greater role in this function. A similar pattern was observed for two-leg RSI, where DL plantar flexion strength and NDL dorsiflexion strength were the strongest contributors, consistent with a propulsion-oriented role for the dominant leg and a stabilization-oriented role for the non-dominant leg.

Previous studies in the literature have also shown that strength training of the ankle flexor muscles resulted in a significant improvement in vertical jump performance [36]. Similarly, it has been emphasized that the strength of the ankle flexor muscles plays a potentially important role in improving the dynamic functions of the lower limb, including both horizontal and vertical jumps [37, 38]. In addition, the literature suggests that the capacity to generate strength at the ankle joint is an important determinant of performance differences in single-leg vertical jumps [2, 3, 14, 39], consistent with findings linking balance, jumping, agility, and sprinting performance in youth athletic populations [40, 41]. The effective energy transfer of the ankle muscle-tendon unit, especially in the transition from eccentric to concentric contraction, is associated with an increase in RSI level [16, 42]. A study by Tourillon et al. [16] showed that the isokinetic strength of the ankle plantar flexor muscles is a fundamental element in the generation of a significant propulsive strength during vertical jumping. However, it has also been suggested that the plantar flexor muscles contribute to rapid acceleration during ballistic movements [42]. It is also consistent with previous studies that ankle kinetics have a marked effect on vertical jumping characteristics [17, 43]. Specifically, the arch of the foot has been shown to function as an effective structure in transferring the mechanical strength generated at the ankle forward [44]. Although other factors such as arch stiffness have also been suggested to influence jump performance [37, 45], our results suggest that dorsiflexion and plantar flexion strength appear to be important contributors of RSI level. While the present findings were obtained in soccer players, similar associations between lower-limb strength/mobility profiles and neuromuscular performance outcomes have also been reported in other team sports. For instance, ankle- and lower-limb-related functional asymmetries have been linked to jump, agility, and change-of-direction performance in basketball and volleyball players, suggesting that the mechanisms underlying ankle contribution to reactive strength performance may extend beyond soccer to other explosive team sports involving repeated unilateral loading [46, 47]. This cross-sport consistency reinforces the potential relevance of ankle strength and mobility training for reactive performance across multiple athletic populations. Notably, although the dominant and non-dominant legs did not differ significantly at the absolute performance level (Supplementary Table S1), their functional contribution patterns to RSI diverged: in the NDL, only dorsiflexion strength was significantly associated with RSI, whereas plantar flexion strength was not, unlike in the DL where both directions contributed. This dissociation between

symmetric absolute performance and asymmetric functional association suggests that the dominant and non-dominant legs may serve mechanically distinct roles -the DL contributing more to propulsive force generation and the NDL to postural stabilization- even in the absence of measurable strength or mobility deficits between limbs.

Exploratory analyses did not reveal significant ROM × strength interaction effects or associations between limb asymmetry and bilateral RSI performance after correction for multiple comparisons. This suggests that, within this cohort, dominant and non-dominant limb characteristics may contribute to RSI relatively independently rather than through synergistic or compensatory mechanisms, although this should be re-examined in larger samples given the exploratory nature of these tests.

Limitations

This study has several limitations that should be acknowledged. First, the sample consisted exclusively of male youth soccer players, which limits the generalizability of the findings to female athletes, different age groups, and other sports disciplines. Second, the cross-sectional design of the study restricts the ability to draw causal inferences between ankle strength, ROM, and reactive strength performance. Third, only ankle-related variables were examined, while other important factors such as hip and knee strength, neuromuscular coordination, and training background were not included, limiting our understanding of multi-joint contributions to RSI performance. Fourth, biological maturation status (e.g., peak height velocity) was not assessed or controlled for in this study, which may have influenced age-related differences, particularly given the youth sample. Fifth, potential measurement limitations related to the hand-held dynamometer, goniometer, and RSI test reliability should be acknowledged, as these tools may introduce measurement error despite standardized protocols. Sixth, although all ROM measurements were performed by the same experienced researcher using a standardized protocol, intra-rater reliability (e.g., intraclass correlation coefficient) was not formally calculated in this study, which should be considered when interpreting the ROM findings. Future studies should consider more diverse samples, including female athletes and participants from different sports, and employ longitudinal or experimental designs to better understand the underlying mechanisms.

Conclusions

Our findings suggest that dorsiflexion and plantar flexion AROM and strength levels of the ankle may be associated with the efficiency of the eccentric-concentric transition process of athletes. In other words, a wider AROM and higher ankle strength may contribute to improved explosive jump performance by optimizing ground contact time. Considering that the dorsiflexion capacity of the DL and the plantar flexibility of the NDL are associated with RSI, flexibility and mobility training may be strategically incorporated into training programs. In addition, the inclusion of exercises targeting DL plantar flexors for single leg performance and NDL dorsiflexion muscles for two leg performance may help to improve RSI values. Overall, ankle joint mechanics appear to be important contributors to movement efficiency and reactive strength performance. Accordingly, in addition to traditional lower extremity strength training, resistance exercises aimed at improving dorsiflexion and plantar flexion strength, as well as mobility and flexibility exercises targeting ankle AROM, may play a potentially important role in enhancing RSI.

Practical applications

The findings of this study provide important practical implications for coaches and practitioners working with youth soccer players. Given the associations between ankle strength, AROM, and reactive strength performance, training programs should incorporate targeted exercises aimed at improving both dorsiflexion and plantar flexion capacities. Specifically, enhancing dorsiflexion AROM of the dominant leg and plantar flexion flexibility of the non-dominant leg may be beneficial for optimizing reactive strength performance.

In addition, strength training interventions focusing on plantar flexor muscles of the dominant leg and dorsiflexor muscles of the non-dominant leg may contribute to improvements in single-leg and bilateral jump performance. Coaches are encouraged to integrate mobility, flexibility, and strength exercises into regular training routines to support efficient stretch–shortening cycle function and reduce potential performance limitations associated with restricted ankle function. It should be noted, however, that ankle AROM and strength explained only a low-to-moderate proportion of RSI variance across models. Accordingly, the training recommendations above should be considered as one component of a broader, multifactorial approach to developing reactive strength, rather than as a primary or standalone determinant of performance.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13102-026-02060-6.

Supplementary Material 1.

Acknowledgements
Not applicable.

Authors' contributions
R.F.K., H.İ.Ç., and S.B. contributed to the conceptualization of the study. R.F.K. and S.B. were responsible for the methodology. R.F.K., E.T., Ö.A., and S.B. contributed to the formal analysis. E.T. and Ö.A. contributed to data curation and investigation. E.T., Ö.A., and S.B. prepared the original draft of the manuscript. All authors contributed to the review and editing of the manuscript. S.B. supervised the study. All authors read and approved the final manuscript.

Funding
This research received no external funding.

Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The study was approved by the Ethics Committee of Recep Tayyip Erdoğan University Social and Human Sciences, date 21/05/2025 and approval number 2025/376. All procedures performed were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments. Written informed consent was obtained from all participants and their legal guardians prior to participation, and permission was granted by the affiliated sports clubs.

Consent for publication
Not applicable, as no individual-level identifiable data are included in this study.

Competing interests
The authors declare no competing interests.

Author details
[1]Department of Coaching Education, Movement and Training Sciences, Faculty of Sports Sciences, Marmara University, İstanbul 34722, Turkey
[2]Department of Sports Management, Faculty of Sports Science, Recep Tayyip Erdogan University, Rize 53020, Türkiye
[3]Department of Physical Education and Sports, Postgraduate Education Institute, Recep Tayyip Erdogan University, Rize 53020, Türkiye
[4]Department of Coaching Education, Movement and Training Sciences, Faculty of Sport Sciences, Istanbul Gelisim University, Avcılar, Istanbul 34310, Türkiye
[5]Department of Coaching Education, Faculty of Sport Sciences, Gumushane University, Gumushane 29100, Türkiye

Received: 3 April 2026 / Accepted: 2 September 2026

References

1. Alexe DI, Čaušević D, Čović N, Rani B, Tohănean DI, Abazović E, et al. The relationship between functional movement quality and speed, agility, and jump performance in elite female youth football players. Sports. 2024;12:214.
2. Mohammadi Nia Samakosh H, Brito JP, Shojaedin SS, Hadadnezhad M, Oliveira R. What does provide better effects on balance, strength, and lower extremity muscle function in professional Male soccer players with chronic ankle instability? Hopping or a balance plus strength intervention? A randomized control study. In: Healthcare. MDPI; 2022. p. 1822.
3. Kayhan RF, Terzi E, Bayrakdaroğlu S, Ceylan Hİ, Morgans R, Nikolaidis PT. Exploring the associations between ankle dorsal and plantar flexion strength, reactive strength index, and balance in elite youth soccer players. Sport Sci Health. 2024;20:1305–16.
4. Kimura M, Ogata Y, Akebi T, Ochi M, Hachisuka K, Saeki S. Development and reliability of a hand-held dynamometer device to measure the ankle dorsiflexor muscle strength. J Rehabil Assist Technol Eng. 2018;5:2055668318802558.
5. Hoveidaei AH, Moradi AR, Nakhostin-Ansari A, Mousavi Nasab MM, Taghavi SP, Eghdami S, et al. Risk factors of ankle sprain in soccer players: a systematic review and meta-analysis. Sports. 2025;13:105.
6. Backman LJ, Danielson P. Low range of ankle dorsiflexion predisposes for patellar tendinopathy in junior elite basketball players: a 1-year prospective study. Am J Sports Med. 2011;39:2626–33.
7. Wilczyński B, Cabaj P, Biały M, Zorena K. Impact of lateral ankle sprains on physical function, range of motion, isometric strength and balance in professional soccer players. BMJ Open Sport Exerc Med. 2024;10:e002293.
8. Clark RA, Humphries B, Hohmann E, Bryant AL. The influence of variable range of motion training on neuromuscular performance and control of external loads. J Strength Conditioning Res. 2011;25:704–11.
9. Pinto RS, Gomes N, Radaelli R, Botton CE, Brown LE, Bottaro M. Effect of range of motion on muscle strength and thickness. J Strength Conditioning Res. 2012;26:2140–5.
10. Jarvis P, Turner A, Read P, Bishop C. Reactive strength index and its associations with measures of physical and sports performance: A systematic review with meta-analysis. Sports Med. 2022;52:301–30.
11. Kayhan RF, Terzi E, Bayrakdaroğlu S, Ceylan Hİ, Bıyıklı T. Isometric shoulder and hip strength impact on throwing velocity and reactive strength. Int J Sports Med. 2024;45:1055–65.
12. Markwick WJ, Bird SP, Tufano JJ, Seitz LB, Haff GG. The intraday reliability of the reactive strength index calculated from a drop jump in professional men's basketball. Int J Sports Physiol Perform. 2015;10:482–8.
13. Werstein KM, Lund RJ. The effects of two stretching protocols on the reactive strength index in female soccer and rugby players. J Strength Conditioning Res. 2012;26:1564–7.
14. Panoutsakopoulos V, Kotzamanidou MC, Giannakos AK, Kollias IA. Relationship of vertical jump performance and ankle joint range of motion: Effect of knee joint angle and handedness in young adult handball players. Sports. 2022;10:86.
15. Nagahara R, Naito H, Miyashiro K, Morin JB, Zushi K. Traditional and ankle-specific vertical jumps as strength-power indicators for maximal sprint acceleration. J Sports Med Phys Fit. 2014;54:691–9.
16. Tourillon R, Six A, Bothorel H, Fourchet F. Are foot posture and morphological deformation associated with ankle plantar flexion isokinetic strength and vertical drop jump kinetics? A principal component analysis. Sports Biomech. 2025;24:2984–98.
17. Cen X, Xu D, Baker JS, Gu Y. Association of arch stiffness with plantar impulse distribution during walking, running, and gait termination. Int J Environ Res Public Health. 2020;17:2090.
18. Kalata M, Hank M, Bujnovsky D, Michalek J, Varjan M, Kunzmann E, et al. Bilateral strength asymmetry in elite youth soccer players: differences between age categories. Symmetry (Basel). 2021;13:1982.
19. Codreanu A-M, Korodi D-A, Lupu N-A, Onciulenco A-V, Vataman A-A, Duse A-O, et al. Multimodal Functional Assessment of Asymmetries in Youth Soccer Players. Study Protoc. 2026;16:876.
20. Zghal F, Rebai H, Colson SS, Samozino P, Rahmani A, Peyrot N, et al. Age-Related Differences in Jumping and Sprinting Performance and Force Production Capacities in Young Soccer Players. Eur J Sport Sci. 2025;25:e12301.
21. Parr J, Winwood K, Hodson-Tole E, Deconinck FJA, Hill JP, Teunissen JW, et al. The main and interactive effects of biological maturity and relative age on physical performance in elite youth soccer players. J Sports Med. 2020;2020:1957636.
22. Tounsi M, Aouichaoui C, Tabka Z, Trabelsi Y. Specific physical performances among male elite youth soccer players: effect of maturity status. J Sports Med Phys Fit. 2021;61:1339–47.
23. Doua N, Marzouki H, Selmi O, Cherni B, Djeddi R, Alexe DI, et al. Comparative effects of reactive and planned agility training on physical performance, internal load and enjoyment in youth soccer players. Sci Rep. 2026;16:11302.
24. Çıkıkcı A, Gülez O, Kayhan RF. Examining the Relationship Between Sprint and Endurance Performances, Distances Covered During the Season, Positional Roles, and Success in Women's Football Players. J Athletic Perform Sports Med. 2024;1:34–43.
25. Cho SY, Myong Y, Park S, Cho M, Kim S. A portable articulated dynamometer for ankle dorsiflexion and plantar flexion strength measurement: A design, validation, and user experience study. Sci Rep. 2023;13:22221.
26. Katoh M. Test-retest reliability of isometric ankle plantar flexion strength measurement performed by a hand-held dynamometer considering fixation: examination of healthy young participants. J Phys Ther Sci. 2022;34:463–6.

27. Schindler JB, Hancock CL, Zifchock RA, Bode VG, Ramsay JW. The intra-and interrater reliability of isometric muscle contractions using a novel fixation system for a hand-held dynamometer. Eur J Sport Sci. 2024;24:1313–8.
28. Spink MJ, Fotoohabadi MR, Menz HB. Foot and ankle strength assessment using hand-held dynamometry: reliability and age-related differences. Gerontology. 2010;56:525–32.
29. Thornton J, Sabah S, Segaren N, Cullen N, Singh D, Goldberg A. Validated method for measuring functional range of motion in patients with ankle arthritis. Foot Ankle Int. 2016;37:868–73.
30. Kenny IC, Caireallán AÓ, Comyns TM. Validation of an electronic jump mat to assess stretch-shortening cycle function. J Strength Conditioning Res. 2012;26:1601–8.
31. Marcoulides KM, Raykov T. Evaluation of variance inflation factors in regression models using latent variable modeling methods. Educ Psychol Meas. 2019;79:874–82.
32. Marshall BM, Moran KA. Which drop jump technique is most effective at enhancing countermovement jump ability,countermovement drop jump or bounce drop jump? J Sports Sci. 2013;31:1368–74.
33. Cerrillo-Sanchis J, Ricart-Luna B, Rodrigo-Mallorca D, Muñoz-Gómez E, Domínguez-Navarro F, Mollà-Casanova S, et al. Relationship between ankle dorsiflexion range of motion and sprinting and jumping ability in young athletes. J Bodyw Mov Ther. 2024;39:43–9.
34. Kipp K, Kiely MT, Giordanelli MD, Malloy PJ, Geiser CF. Biomechanical determinants of the reactive strength index during drop jumps. Int J Sports Physiol Perform. 2018;13:44–9.
35. Massey CD, Vincent J, Maneval M, Johnson JT. Influence of range of motion in resistance training in women: early phase adaptations. J Strength Conditioning Res. 2005;19:409–11.
36. Jung D-Y, Koh E-K, Kwon O-Y. Effect of foot orthoses and short-foot exercise on the cross-sectional area of the abductor hallucis muscle in subjects with pes planus: a randomized controlled trial. J Back Musculoskelet Rehabil. 2011;24:225–31.
37. Smith RE, Lichtwark GA, Kelly LA. Flexor digitorum brevis utilizes elastic strain energy to contribute to both work generation and energy absorption at the foot. J Exp Biol. 2022;225:jeb243792.
38. Suzuki Y, Murata M. Comparison of translational momentum and mechanical energy produced by lower limb muscles between horizontal and vertical jumps–a computer simulation study. Int J Sport Health Sci. 2020;18:207–14.
39. Kozinc Ž, Šarabon N. Measurements of lower-limb isometric single-joint maximal voluntary torque and rate of torque development capacity offer limited insight into vertical jumping performance. Meas Phys Educ Exerc Sci. 2022;26:15–26.
40. Jouira G, Rebai H, Alexe DI, Sahli S. Postural balance in boys with intellectual disabilities who participate in soccer training. Pediatr Exerc Sci. 2024;37:37–45.
41. Jouira G, Alexe DI, Tohănean DI, Alexe CI, Tomozei RA, Sahli S. The relationship between dynamic balance, jumping ability, and agility with 100 m sprinting performance in athletes with intellectual disabilities. Sports. 2024;12:58.
42. Pandy MG, Lai AKM, Schache AG, Lin Y. How muscles maximize performance in accelerated sprinting. Scand J Med Sci Sports. 2021;31:1882–96.
43. Grozier CD, Cagle GK, Pantone L, Rank KB, Wilson SJ, Harry JR, et al. Effects of medial longitudinal arch flexibility on propulsion kinetics during drop vertical jumps. J Biomech. 2021;118:110322.
44. Takahashi KZ, Gross MT, Van Werkhoven H, Piazza SJ, Sawicki GS. Adding stiffness to the foot modulates soleus force-velocity behaviour during human walking. Sci Rep. 2016;6:29870.
45. Kelly LA, Farris DJ, Cresswell AG, Lichtwark GA. Intrinsic foot muscles contribute to elastic energy storage and return in the human foot. J Appl Physiol. 2019;126:231–8.
46. Yılmaz AK, Akgün S, Salkılıç EK, Anıl B, Akdemir E, Aktaş B, et al. The relationship between agility and lower extremity strength in female basketball players. BMC Sports Sci Med Rehabil. 2025;17:198.
47. Akgün S, Atıcı M, Bayrakdar A, Salkılıç EK, Anıl B, Akdemir E, et al. Relationship between lower limb neuromuscular and functional performance tests and sprint and agility in female volleyball players: a cross-sectional study. Front Physiol. 2026;17:1827693.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.